Graph-Smooth Sparse Orthogonal PCA via Alternating Manifold and Graph-Coupled Proximal Updates

Wang Zhoufu

Abstract

We study sparse, graph-smooth, explicitly orthogonal multi-component PCA through a split-variable formulation that couples an orthogonal loading matrix with a sparse graph-regularized proxy. We propose a concrete alternating solver consisting of a Stiefel manifold update for the orthogonal factor and a proximal-gradient update for the sparse graph-smooth factor. We evaluate the method on synthetic graph families using shared evaluation metrics (support F1, graph smoothness norm on the true graph, and shared explained variance). In our synthetic experiments, graph regularization yields smoother solutions on the true graph, while its effect on support recovery is topology-dependent: on chain graphs the method is competitive on recovery and smoother on the true graph, whereas on SBM graphs stronger regularization induces a recovery–smoothness tradeoff and light regularization can improve smoothness with limited recovery loss. In the graph-aligned chain setting with disjoint contiguous supports and scattered high-variance decoys, graph regularization exhibits a repeatable low-to-moderate (λ_2) operating regime in which smoothness improves substantially relative to SparseNoGraph with no clear support F1 degradation; robustness sweeps over λ_1 and ρ show that this regime is consistent across tested settings, although its width is tuning-dependent. The study is synthetic-first and limited to chain and SBM topologies; broader theoretical guarantees and real-data validation remain future work.

1 Introduction

Principal component analysis (PCA) is a foundational technique for dimensionality reduction, exploratory analysis, and low-dimensional representation learning. Its appeal stems from its simplicity and its optimality for variance maximization under orthogonality constraints. However, classical PCA typically produces dense loading vectors, so that each principal component depends on nearly all input variables. In modern high-dimensional applications such as genomics, neuroimaging, and structured sensing, these dense loadings can be difficult to interpret and of limited use for scientific discovery or feature selection [19, 5].

Sparse principal component analysis (SPCA) addresses this limitation by encouraging principal components to depend on only a small subset of variables. Over the past two decades, SPCA has been studied through several complementary formulations, including regression-based relaxations, power-method variants, and nonsmooth constrained optimization approaches [19, 11, 5]. More recent work has emphasized the importance of handling sparsity and orthogonality in a principled manner, rather than relying solely on heuristic thresholding or post hoc orthogonalization. In particular, manifold-based formulations such as alternating manifold proximal-gradient methods show that sparse multi-component estimation can be treated directly as a nonsmooth optimization problem with orthogonality constraints, yielding stronger algorithmic structure and convergence guarantees than purely heuristic alternatives [5, 4].

Despite these advances, sparsity alone is often not enough. In many high-dimensional domains, features are not exchangeable but are linked through known relational structure. Genes participate in pathways and interaction networks, voxels exhibit spatial or functional connectivity, and pixels or spectral bands obey spatial adjacency and smoothness patterns. When such prior structure is available, it is natural to ask whether the extracted principal components should also respect it. A large literature on graph-aware and graph-regularized PCA studies how incorporating structural priors through graph-based penalties or low-frequency graph constraints can improve robustness and structural fidelity, especially when the signal is aligned with the underlying graph [2]. At the same time, this literature also makes clear that graph-aware PCA is not a single unified model class: some methods emphasize low-frequency graph consistency, some include explicit sparsification, and others target application-specific clustering or representation objectives rather than sparse orthogonal component extraction [2, 9, 17, 15].

A second challenge concerns the extraction of multiple components. In classical PCA, orthogonality follows directly from eigendecomposition. In sparse settings, however, orthogonality is no longer automatic, and sequential deflation can produce components that are only approximately orthogonal or that degrade cumulatively in quality. This has motivated a continued line of work on simultaneous sparse orthogonal extraction, including recent efforts that explicitly revisit orthogonal SPCA as an active optimization problem rather than a solved

issue [5, 1, 6]. These developments suggest that, when multiple interpretable components are required, joint extraction under explicit orthogonality constraints remains preferable to treating sparsity as a purely post-processing step.

There is also a broader methodological backdrop. Beyond SPCA itself, recent work on nonsmooth and structured nonsmooth optimization over Riemannian submanifolds has substantially expanded the toolkit available for nonconvex problems with manifold constraints [12, 4]. Moreover, adjacent problems such as sparse tensor canonical correlation analysis already combine Laplacian regularization, sparse penalties, manifold-based optimization, and modern inner solvers with convergence analysis [18]. Consequently, it is no longer accurate to position graph-coupled nonsmooth manifold optimization as an entirely unexplored conceptual territory. Rather, the more defensible and practically meaningful question is whether these ideas can be brought together in a coherent and effective framework for *PCA proper*, with an explicit focus on sparse multi-component loading estimation under feature-graph smoothness and simultaneous orthogonality.

This paper studies that question. We consider a formulation that combines three desiderata: sparse loadings, smoothness with respect to a known feature graph, and joint orthogonality across multiple extracted components. Our goal is not to claim that graph-aware PCA, graph+sparsity PCA, or nonsmooth manifold optimization are individually new. Instead, we focus on a comparatively underexplored intersection of these themes: *feature-graph Laplacian regularization, sparse multi-component loadings, and explicit simultaneous orthogonality within a modern manifold/proximal optimization framework for PCA*. This positioning is narrower than a broad novelty claim, but it better reflects the current state of the literature and the genuine methodological gap that remains [2, 9, 17, 5].

Concretely, we introduce a graph-smooth sparse orthogonal PCA formulation in which orthogonality is handled through a Stiefel-constrained variable and structured sparse estimation is handled through a graph-regularized Euclidean variable coupled by a quadratic penalty. This split allows us to separate the geometry of orthogonality from the graph-coupled sparsity structure, while still optimizing the two jointly. The resulting viewpoint is particularly attractive in settings where interpretability is not just a matter of selecting a few variables, but of selecting variables whose pattern is coherent with known domain structure.

Our study is guided by two central questions. First, when does feature-graph regularization preserve recovery while improving structural coherence of sparse components? Second, what tradeoff does graph regularization induce between explained variance, graph consistency, and computational cost? These questions are important because graph-based regularization can help when the prior graph is informative, but may oversmooth the loadings or reduce variance capture when the graph is misspecified. Accordingly, the paper emphasizes not only model design and optimization, but also the empirical circumstances under which graph priors are beneficial.

Contributions. The main contributions of this work are as follows.

- Graph regularization consistently reduces the chosen graph-smoothness metric on the true graph, while its recovery impact depends on topology.
- On chain graphs, recovery remains competitive with sparse baselines while smoothness improves; on SBM graphs, stronger regularization induces a clearer recovery–smoothness tradeoff.
- In a graph-aligned chain setting with disjoint contiguous supports and scattered high-variance decoys, we identify a repeatable low-to-moderate (λ_2) operating regime with strong smoothness gains and no detectable recovery degradation relative to SparseNoGraph; this regime is consistent across tested λ_1 and ρ settings, though its width is tuning-dependent.
- We provide a concrete split-variable solver that combines explicit orthogonality, sparsity, and graph smoothness within a single executable framework.

Taken together, these contributions position the paper as a targeted integration of ideas that already exist in neighboring literatures, but whose precise combination for graph-smooth sparse orthogonal PCA remains comparatively underdeveloped. Our aim is therefore not to argue that every ingredient is new, but to show that their combination leads to a useful and well-motivated framework for structured dimensionality reduction in graph-informed high-dimensional settings.

2 Related Work

2.1 Sparse PCA and Orthogonal Component Extraction

Sparse principal component analysis has been studied through regression-based relaxations, truncated power methods, penalized matrix decomposition, semidefinite relaxations, and nonsmooth constrained optimization [19, 16, 8, 11, 13]. A central theme in this literature is that sparse loadings improve interpretability but complicate both optimization and the extraction of multiple components.

A particularly important line of work formulates SPCA directly as a nonsmooth optimization problem over orthogonality-constrained variables. Chen et al. proposed the alternating manifold proximal gradient framework (A-ManPG) for sparse PCA and sparse CCA, showing

that sparse loadings and nonconvex orthogonality constraints can be handled within a unified manifold-based optimization scheme with convergence guarantees [5]. More broadly, the ManPG literature establishes a practical optimization template for smooth-plus-nonsmooth objectives on the Stiefel manifold [3, 4].

Orthogonal sparse extraction also remains an active topic rather than a closed one. GeoSPCA emphasizes simultaneous extraction over deflation for orthogonal sparse components [1], and recent work on certifiably optimal orthogonal SPCA further underscores ongoing interest in jointly enforcing sparsity, orthogonality, and strong optimality properties [6]. These developments motivate explicit joint orthogonality rather than treating it as an afterthought.

2.2 Graph-Aware PCA and Graph-Regularized Dimensionality Reduction

A parallel literature incorporates structural priors into PCA through graphs or Laplacian-type regularization. These methods are motivated by the observation that in many high-dimensional applications, features are not exchangeable but related through biological, spatial, or network structure. Graph-aware PCA methods use this prior information to bias principal components toward graph-consistent directions, often by suppressing high-frequency variation and preserving graph-coherent low-frequency signals [2, 7, 14].

Recent work makes it clear that graph-regularized PCA cannot simply be characterized as producing dense loadings. In particular, GR-PCA incorporates a learned sparse precision graph and regularizes PCA toward the low-frequency Fourier modes of the induced graph Laplacian; the method also evaluates sparsity-related behavior explicitly and reports improved structural fidelity, lower Laplacian energy, and competitive out-of-sample reconstruction under graph-aligned signals [2]. This weakens any broad claim that graph-aware PCA is predominantly dense or uninterpretable.

At the same time, graph-aware PCA does not fully eliminate the gap relevant here. Many graph-regularized variants focus primarily on graph consistency, robustness, clustering structure, or low-frequency filtering, rather than on simultaneous sparse multi-component orthogonality in a single PCA-specific framework. Accordingly, the relevant distinction is no longer whether graph-aware PCA exists, but whether it addresses the specific combination of sparse loadings, feature-graph smoothness, and explicit joint orthogonality considered here.

2.3 Graph + Sparsity PCA Variants

Older and application-driven papers already show that graph structure and sparsity have been combined within PCA-like models. For example, gLSPCA integrates a graph Laplacian term and a sparsity constraint into PCA for gene-expression analysis [9], while GDSPCA combines graph Laplacian regularization with double sparsity constraints to improve interpretability and exploit internal geometry [17]. These papers are direct counterexamples to any claim that graph+sparsity PCA is absent from the literature.

However, these graph+sparsity PCA methods also illustrate why the gap must be phrased carefully. They are often application-specific, may target particular geometric or clustering objectives, and do not necessarily adopt the same optimization language as modern manifold proximal methods for joint orthogonal extraction. Recent applied work such as RSSPCA reinforces the same point: smooth sparse PCA variants continue to emerge in domain-specific settings, but they do not by themselves close the gap for feature-graph Laplacian regularization with explicit simultaneous orthogonal extraction in PCA proper [15].

2.4 Structured Sparse PCA

Structured sparse PCA extends plain ℓ_1-based sparsity by encouraging supports that follow meaningful patterns such as groups, trees, or contiguous regions. Jenatton, Obozinski, and Bach showed that structured regularization can impose higher-order information on the support of loading vectors and can improve interpretability relative to unstructured sparse PCA [10]. This literature is highly relevant because it demonstrates that sparsity alone is often too weak a prior.

Nevertheless, structured sparse PCA in this sense is not identical to feature-graph smoothness. Structured sparsity typically constrains the support geometry through groups or shapes, whereas graph-Laplacian regularization encourages neighboring or strongly connected features to take similar values. These are related but distinct inductive biases, and the existence of structured sparse PCA does not by itself resolve the specific question of how to jointly impose graph smoothness, sparsity, and simultaneous orthogonality in multi-component PCA.

2.5 Structured Nonsmooth Optimization on Manifolds

The optimization backdrop for this problem has evolved substantially. Beyond the sparse-PCA-specific literature, recent work on nonsmooth and structured nonsmooth optimization over Riemannian submanifolds shows that manifold-based methods now handle a broader class of composite objectives than earlier sparse-only formulations [12, 4]. Recent adjacent work also combines sparse regularization, Laplacian regularization, manifold proximal updates, semismooth Newton subproblem solvers, and convergence analysis in related problems such as sparse tensor CCA [18]. That literature does not solve our PCA formulation directly, but it does weaken any

broad claim that the optimization ingredients themselves are unprecedented.

2.6 Positioning of the Present Work

Graph-aware PCA and graph-regularized sparse PCA have been studied in multiple forms, and we do not claim novelty in combining graph smoothness with sparsity. The specific gap we target is the joint extraction of multiple explicitly orthogonal components under a split-variable formulation with a concrete, reproducible solver. Our work is positioned as a focused intersection of sparse PCA, graph-regularized PCA, and orthogonality-preserving optimization, rather than as a first instance of any single ingredient.

3 Problem Setup

Let $X \in \mathbb{R}^{n\times p}$ be a centered data matrix and let

$$\widehat{\Sigma} = \frac{1}{n} X^\top X \tag{1}$$

denote the sample covariance matrix. Let $G = (\mathcal{V}, \mathcal{E}, W)$ be a weighted undirected feature graph on the p variables, with adjacency matrix W, degree matrix D, and graph Laplacian

$$L = D - W. \tag{2}$$

We assume $L \succeq 0$. We use $\|M\|_1$ for the entrywise ℓ_1 norm, $\mathrm{soft}(Z, \tau)$ for entrywise soft-thresholding at threshold τ, and $\mathrm{sym}(M) = (M + M^\top)/2$ for the symmetric part of a matrix. Shared explained variance is computed after QR-based orthonormalization of the loading matrix.

Our goal is to extract r principal components with loading structure that is simultaneously sparse, graph-smooth, and orthogonal. To separate the geometry of orthogonality from the structured regularization, we introduce two variables $A, B \in \mathbb{R}^{p\times r}$:

- A carries the orthogonality constraint $A^\top A = I_r$,
- B carries sparsity and graph smoothness.

We denote the Stiefel manifold by

$$\mathrm{St}(p, r) = \{A \in \mathbb{R}^{p\times r} : A^\top A = I_r\}. \tag{3}$$

4 Proposed Formulation

We study the split formulation

$$\begin{aligned} \min_{A\in\mathrm{St}(p,r),\, B\in\mathbb{R}^{p\times r}} \quad & -\operatorname{tr}(A^\top \widehat{\Sigma} A) + \lambda_1 \|B\|_1 \\ & + \lambda_2 \operatorname{tr}(B^\top L B) + \frac{\rho}{2}\|A - B\|_F^2. \end{aligned} \tag{4}$$

The first term promotes variance capture; the ℓ_1 term promotes sparse loadings; the Laplacian term promotes graph smoothness; and the quadratic coupling term forces the orthogonal variable A and the structured loading variable B to remain close.

Penalty coupling. The coupling term $\frac{\rho}{2}\|A - B\|_F^2$ is a fixed-ρ quadratic penalty that enforces approximate consensus between A and B; we do not use dual variables or ADMM-style updates.

Why split variables? Directly optimizing a sparse graph-regularized objective under orthogonality constraints is difficult because the orthogonality constraint is nonconvex while the graph-sparsity regularizer is nonsmooth and coupled across features. The split formulation isolates these effects: A is updated on the Stiefel manifold, while B is updated in Euclidean space under structured regularization.

We leave alternative sparsity structures to future work to keep the formulation fixed to entrywise ℓ_1 on B.

5 Optimization Algorithm

We solve the split-variable objective by alternating between two steps. The A-step updates an orthogonal loading matrix on the Stiefel manifold using a Riemannian gradient step followed by QR-based retraction. The B-step updates the sparse graph-smooth proxy with proximal gradient, treating the Laplacian quadratic and coupling term as the smooth part and the ℓ_1 penalty as the nonsmooth part. This single solver path provides a minimal, reproducible baseline for the empirical study; more advanced variants remain future work.

5.1 A-Step: Manifold Update

With B fixed, the A-subproblem is

$$\min_{A\in\mathrm{St}(p,r)} -\operatorname{tr}(A^\top \widehat{\Sigma} A) + \frac{\rho}{2}\|A - B\|_F^2. \tag{5}$$

Its Euclidean gradient is

$$\nabla_A f(A; B) = -2\widehat{\Sigma} A + \rho(A - B). \tag{6}$$

The corresponding Riemannian gradient on $\mathrm{St}(p, r)$ is

$$\operatorname{grad} f(A; B) = \nabla_A f(A; B) - A\, \mathrm{sym}\big(A^\top \nabla_A f(A; B)\big). \tag{7}$$

We take a fixed-step manifold descent update with step size η_A and retract back to the Stiefel manifold using QR retraction.

5.2 B-Step: Graph-Coupled Sparse Update

With A fixed, the B-subproblem is

$$\min_{B\in\mathbb{R}^{p\times r}} \lambda_1 \|B\|_1 + \lambda_2 \operatorname{tr}(B^\top L B) + \frac{\rho}{2}\|A - B\|_F^2. \tag{8}$$

This subproblem is convex in B because $L \succeq 0$. We use a proximal-gradient update that treats the Laplacian

Algorithm 1 Alternating Manifold + Graph-Coupled Sparse Updates

Require: Centered data X, graph Laplacian L, parameters $\lambda_1, \lambda_2, \rho$, rank r

Initialize $A^{(0)} \in \mathrm{St}(p, r)$, $B^{(0)} \in \mathbb{R}^{p \times r}$
for $t = 0, 1, \ldots, T-1$ **do**
  Update $A^{(t+1)}$ with one Riemannian gradient step and QR retraction
  Update $B^{(t+1)}$ with one proximal-gradient step and soft-thresholding
  if stopping criterion satisfied **then**
    break
  end if
end for
return $(A^{(T)}, B^{(T)})$

quadratic and coupling term as the smooth part and the ℓ_1 penalty as the nonsmooth part. Define

$$g(B) = \lambda_2 \operatorname{tr}(B^\top L B) + \frac{\rho}{2}\|A - B\|_F^2, \qquad h(B) = \lambda_1 \|B\|_1.$$

Then

$$\nabla g(B) = 2\lambda_2 L B + \rho(B - A).$$

With step size $\eta_B = 1/(2\lambda_2\|L\|_2 + \rho)$, the update is

$$B^{(t+1)} = \operatorname{soft}\big(B^{(t)} - \eta_B \nabla g(B^{(t)}), \eta_B \lambda_1\big).$$

5.3 Alternating Scheme

Algorithm 1 summarizes the outer loop.

Complexity. The cost of the A-step is dominated by matrix multiplications and the chosen retraction. The B-step is typically the computational bottleneck because graph regularization introduces feature coupling and removes the closed-form separability available in plain soft-thresholding. When L is sparse, matrix-vector operations remain efficient; dense or ill-conditioned graph operators can substantially increase runtime.

6 Optimization Discussion

The objective in (4) is nonconvex because it combines a Stiefel constraint on A with a quadratic coupling term between A and B. For fixed A, however, the B-subproblem is convex because it is the sum of a convex quadratic term and the entrywise ℓ_1 norm.

6.1 Updates Used

The implemented solver alternates between two explicit updates. The A-step takes one retracted Riemannian gradient step on $\mathrm{St}(p, r)$ using the Euclidean gradient from the manifold objective and QR retraction back to the manifold. The B-step takes one proximal-gradient step, with the smooth part given by the Laplacian quadratic and coupling term and the nonsmooth part given by $\lambda_1\|B\|_1$.

The step size for the A-step is fixed at $\eta_A = 0.05$. This value is conservative relative to the Lipschitz bound for the smooth A-subproblem (on the order of $2\|\widehat{\Sigma}\|_2 + \rho$), and was not tuned to optimize evaluation metrics. The B-step uses

$$\eta_B = \frac{1}{2\lambda_2\|L\|_2 + \rho},$$

which matches the Lipschitz constant of the smooth part in the Euclidean update. We compute $\|L\|_2$ as the largest eigenvalue of L (via `eigvalsh`).

6.2 Stopping Criteria

The outer loop stops when any of the following is satisfied (defaults used in all reported runs):

- relative objective change $\leq 10^{-6}$,
- coupling gap $\|A - B\|_F / \max(1, \|A\|_F) \leq 10^{-4}$,
- orthogonality defect $\|A^\top A - I_r\|_F \leq 10^{-6}$,
- maximum iteration count $T = 200$ is reached.

This section is intentionally descriptive rather than a proof development: we do not provide a formal convergence proof here, and formal analysis is out of scope for this manuscript.

7 Experimental Setup

We evaluate only the executed synthetic study. The comparison set is limited to PCA, A-ManPG, SparseNoGraph, and the Proposed method. SparseNoGraph is the in-family ablation with $\lambda_2 = 0$, while A-ManPG is the minimal external sparse orthogonal baseline without graph coupling. The graph families are chain and SBM. Main robustness results use seeds 0–4, sweep experiments use seeds 0–2, and the graph-aligned decoy study uses seeds 0–19. Shared evaluation metrics only are reported: support F1, graph smoothness norm on the true graph, shared explained variance, and support connectivity.

Metric mapping and definitions. Support metrics (precision/recall/F1 on the union and per component) are computed on B after component matching to the true loadings; we use a Hungarian matching and sign correction before thresholding. Graph smoothness is reported as the normalized Laplacian energy

$$\mathrm{Smoothness}(B; L) = \frac{\operatorname{tr}(B^\top L B)}{\operatorname{tr}(B^\top B)},$$

computed on the true graph Laplacian. Shared explained variance is computed on an orthonormalized loading matrix: for Proposed and SparseNoGraph we use $Q = \mathrm{qr}(B)$,

for A-ManPG we use $Q = \mathrm{qr}(A)$, and for PCA we use the top-r eigenvectors of $\widehat{\Sigma}$. Support connectivity is computed on the union support mask (entries of B exceeding 10^{-8}) and reported as the number of connected components and the largest-component ratio.

A-ManPG settings. We use the public A-ManPG baseline with step size $\eta_A = 0.05$, maximum iterations 200, and tolerances $\mathrm{tol}_{\mathrm{obj}} = 10^{-6}$ and $\mathrm{tol}_{\mathrm{orth}} = 10^{-6}$, initialized from the PCA solution.

7.1 Main Runs

The main runs use the following hyperparameters unless otherwise stated:

- $\lambda_1 = 0.1$,
- $\lambda_2 = 0.1$,
- $\rho = 5.0$,
- $\eta_A = 0.05$.

7.2 Graph-Aligned Chain Setting

To test when the graph prior is genuinely helpful, we introduce a graph-aligned chain regime. Each component has a disjoint contiguous support block, evenly spaced along the chain with clear gaps. We add scattered high-variance decoy features sampled from non-support nodes, with three decoy intensities (counts $\{6, 12, 18\}$ and variance factors $\{1.5, 2.0, 2.5\}$ relative to σ^2 at $p = 200$). This construction creates a setting where graph-unaware sparsity can be drawn toward isolated high-variance nodes, while graph regularization should favor coherent contiguous supports.

7.3 Multi-Seed λ_2 Sweeps

We report multi-seed sweeps over $\lambda_2 \in \{0.0, 0.05, 0.1, 0.2, 0.5\}$ for SBM (mandatory) and for chain (recommended), using seeds 0–2. We plot mean curves with uncertainty bands and overlay A-ManPG as a horizontal reference line (mean across the same seed set).

8 Results

Overview. We report synthetic results using shared evaluation metrics across all methods: union support F1, graph smoothness norm on the true graph, shared explained variance (after QR orthonormalization), and a support-connectivity diagnostic (largest connected-component ratio). Shared explained variance is an evaluation metric and is not the training objective for any method. We do not compare raw objective values across methods because they optimize different objectives. Results are presented for chain and SBM graphs across five seeds (0–4), for multi-seed λ_2 sweeps (seeds 0–2), and for the graph-aligned decoy study (seeds 0–19). The comparison set includes PCA, A-ManPG, SparseNoGraph, and the Proposed method.

Table 1: Chain robustness (seeds 0–4). Mean ± std for shared evaluation metrics.

Method	Support F1	Smoothness (norm)	Shared explained var
PCA	0.261 ± 0.000	2.099 ± 0.484	6.225 ± 0.406
A-ManPG	0.914 ± 0.078	2.111 ± 0.483	6.122 ± 0.408
SparseNoGraph	0.950 ± 0.033	2.118 ± 0.476	6.092 ± 0.413
Proposed	0.927 ± 0.019	1.871 ± 0.465	6.059 ± 0.415

Table 2: SBM robustness (seeds 0–4). Mean ± std for shared evaluation metrics.

Method	Support F1	Smoothness (norm)	Shared explained var
PCA	0.261 ± 0.000	17.085 ± 0.974	6.225 ± 0.406
A-ManPG	0.914 ± 0.078	17.134 ± 1.008	6.122 ± 0.408
SparseNoGraph	0.950 ± 0.033	17.166 ± 1.035	6.092 ± 0.413
Proposed	0.744 ± 0.051	15.764 ± 1.104	6.003 ± 0.384

8.1 Chain Robustness (Seeds 0–4)

Across seeds, PCA is consistently worst on support recovery. A-ManPG improves substantially over PCA but remains below the sparse baselines on mean support F1. SparseNoGraph achieves the highest mean support F1, while the Proposed method attains slightly lower support F1 but achieves a lower graph smoothness norm on the true chain graph. Shared explained variance is similar across the non-PCA methods and lower than PCA, as expected for sparse baselines. This pattern indicates that on chain graphs, graph regularization improves structural coherence while remaining competitive on recovery.

8.2 SBM Robustness (Seeds 0–4)

On SBM graphs, PCA again performs worst on support recovery. A-ManPG again improves over PCA but does not match the sparse baselines on support recovery. SparseNoGraph attains substantially higher support F1 than the Proposed method, while the Proposed method is consistently smoother on the true SBM graph (lower graph smoothness norm). Shared explained variance remains comparable across the sparse methods. These results show a stable recovery–smoothness tradeoff on SBM: stronger graph regularization improves structural smoothness but can reduce recovery accuracy.

8.3 Graph-Aligned Chain Setting (Disjoint Supports + Scattered Decoys)

Figure 1 summarizes the high-decoy graph-aligned setting. Robustness sweeps over λ_1 and ρ show that the low-to-moderate (λ_2) operating regime is repeatable across all

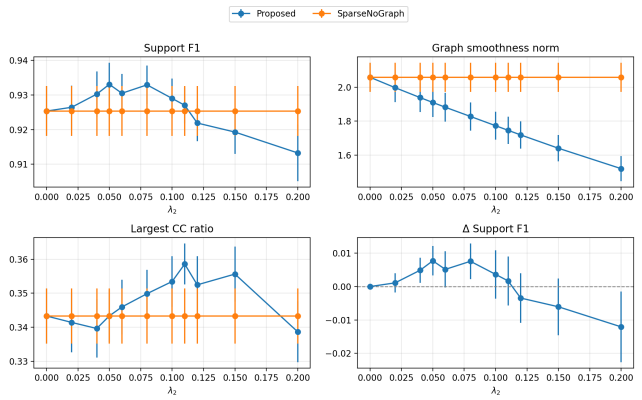

Figure 1: High-decoy graph-aligned setting (20 seeds, mean ± SEM). In the high-decoy regime, the proposed method shows a low-to-moderate (λ_2) operating regime with strong reductions in graph smoothness norm relative to SparseNoGraph and no clear support F1 penalty. The ΔSupport F1 panel (Proposed − SparseNoGraph) stays near zero across the regime, supporting the preservation claim. Across the same regime, the proposed method exceeds A-ManPG on support F1, graph smoothness norm, and connectivity. Larger λ_2 values oversmooth and eventually reduce recovery.

Table 3: Graph-aligned chain, high-decoy representative setting ($\lambda_2 = 0.05$, 20 seeds). Mean ± std for shared metrics.

Method	Support F1	Smoothness (norm)	Largest CC ratio
PCA	0.261 ± 0.000	2.045 ± 0.389	1.000 ± 0.000
A-ManPG	0.807 ± 0.050	2.054 ± 0.389	0.264 ± 0.028
SparseNoGraph	0.925 ± 0.032	2.058 ± 0.382	0.343 ± 0.036
Proposed	0.933 ± 0.028	1.910 ± 0.379	0.343 ± 0.034

9 tested configurations. Under the default configuration this regime is centered in the previously identified low-(λ_2) region, while the interval broadens or narrows with tuning; in particular, it narrows for $\lambda_1 = 0.05$ and $\rho \in \{5, 10\}$. Table 3 reports a representative $\lambda_2 = 0.05$ setting (20 seeds). Figure 2 provides the medium-decoy supporting view, which follows a similar but weaker pattern.

8.4 Multi-Seed λ_2 Sweeps

Sweeping $\lambda_2 \in \{0.0, 0.05, 0.1, 0.2, 0.5\}$ across seeds (0–2) reveals stable smoothness–recovery tradeoffs with topology-dependent sensitivity. On SBM, stronger regularization steadily reduces support F1 while improving smoothness; light regularization improves smoothness with limited recovery loss. On chain graphs, the same sweep shows a milder recovery penalty for comparable smoothness gains. In all sweeps, A-ManPG provides a fixed sparse baseline reference line.

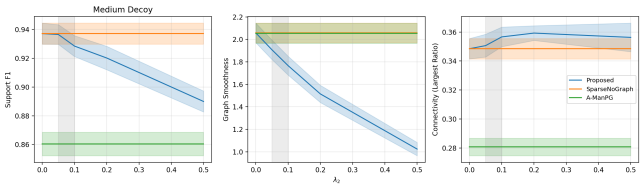

Figure 2: Medium-decoy graph-aligned setting (20 seeds, mean ± SEM). The medium-decoy regime shows a similar but weaker pattern: low-to-moderate graph regularization improves graph smoothness norm while keeping support recovery near the sparse non-graph baseline over a limited λ_2 interval. The benefit is more modest than in the high-decoy setting, indicating that the usefulness of graph regularization depends on distractor strength.

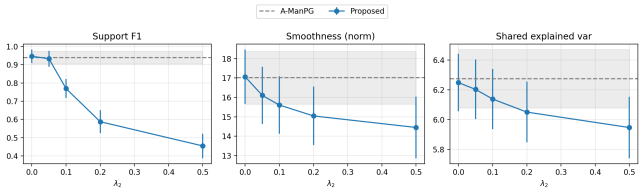

Figure 3: SBM λ_2 sweep (seeds 0–2). Curves show mean ± std across seeds; A-ManPG is a horizontal reference line.

9 Discussion and Limitations

These experiments establish a focused empirical message: in our synthetic experiments, graph regularization yields smoother solutions on the true graph for both chain and SBM settings, but its effect on support recovery is topology-dependent. On chain graphs, the proposed method remains competitive on recovery while producing smoother supports, consistent with the intuition that chain structure aligns well with Laplacian smoothness. On SBM graphs, the graph prior enforces smoother structure at the cost of recovery accuracy, yielding a stable tradeoff rather than a contradiction. In the graph-aligned chain setting with disjoint contiguous supports and scattered high-variance decoys, this operating regime yields substantial smoothness gains relative to SparseNoGraph with no clear support F1 degradation; robustness sweeps over λ_1 and ρ show that the regime is repeatable across tested settings, although its width is tuning-dependent.

The chain setting aligns naturally with local smoothness, whereas the SBM setting emphasizes block/community structure; as a result, the same smoothness penalty can act as a mild regularizer in one topology and as an oversmoothing force in another. The λ_2 sweep on SBM makes this explicit: mild regularization yields smoother solutions with limited recovery loss, whereas stronger regularization oversmooths and reduces recovery. These observations imply that λ_2 must be tuned with respect to graph topology rather than assumed to be universally beneficial.

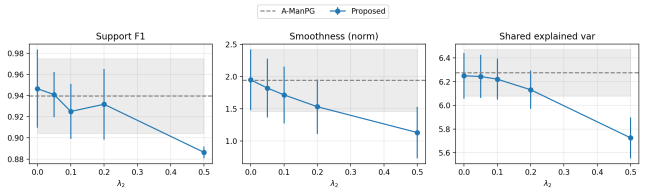

Figure 4: Chain λ_2 sweep (seeds 0–2). Curves show mean $\pm$ std across seeds; A-ManPG is a horizontal reference line.

A small robustness sweep over λ_1 and ρ shows that the operating regime is not confined to a single default configuration: it appears across all tested settings, with broader intervals in most cases and narrower intervals when $\lambda_1 = 0.05$ and $\rho \in \{5, 10\}$. One plausible explanation is that lower λ_1 weakens sparsity pressure and admits more diffuse supports, while weaker coupling (smaller ρ) relaxes consensus between A and B, making it easier for smoothing to spread beyond the true support. We do not claim that graph regularization universally improves support F1 over SparseNoGraph; rather, it delivers structural gains with limited or no recovery loss in graph-aligned regimes and induces a recovery–smoothness tradeoff elsewhere.

The main limitation of the current evidence is that it remains synthetic and focuses on two graph families with a single data generator. This is appropriate for a focused empirical study, whose goal is to establish a disciplined method and a clear empirical tradeoff. The results demonstrate that structural priors can improve graph coherence, but they also show that regularization strength must be tuned to topology. If the graph-aligned setting fails to preserve recovery, the correct conclusion is a tradeoff characterization rather than a recovery-improving claim.

10 Conclusion

We present synthetic evidence that graph-regularized sparse orthogonal PCA improves structural coherence while inducing a topology-dependent tradeoff with support recovery. On chain graphs, the proposed method is competitive on recovery and yields smoother supports; on SBM graphs, stronger graph regularization improves smoothness but reduces recovery accuracy. In the graph-aligned chain setting with disjoint contiguous supports and scattered high-variance decoys, a stable low-to-moderate (λ_2) operating regime delivers substantial smoothness gains relative to SparseNoGraph with no clear support F1 degradation; robustness sweeps over λ_1 and ρ show that this regime persists across all tested settings, although its width is tuning-dependent. These findings highlight the need for topology-aware, tuning-sensitive λ_2 selection and support a modest, defensible claim: graph regularization improves structural coherence, and its recovery impact depends on topology and regularization strength.

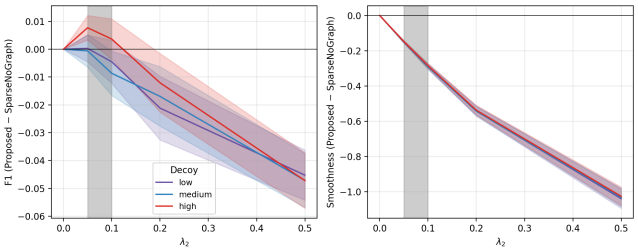

Figure 5: Difference summary relative to SparseNoGraph (20 seeds, mean $\pm$ SEM). Proposed minus SparseNoGraph across decoy intensities. In the high-decoy setting, a low-to-moderate λ_2 region yields strong graph smoothness gains with no clear support F1 penalty; outside this region, larger λ_2 values reduce recovery. Medium-decoy behavior is directionally similar but weaker.

A Supplementary Figures

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